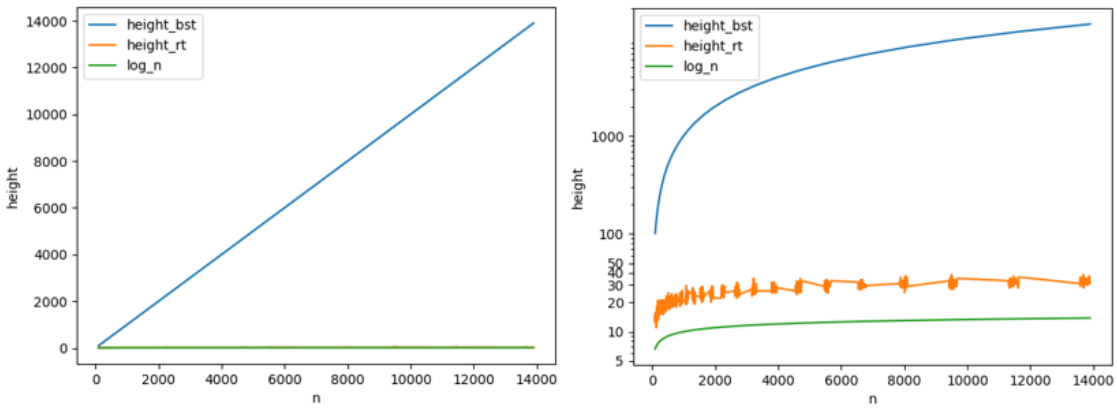


Figure 3.19 The height of a random treap versus a BST, after inserting the sequence `0, 1, …, n-1`. The charts show the same data, on the left with linear scale and on the right using logarithmic scale for heights.

As expected, for BSTs the height is always not just linear in the number of nodes, but equal to the number of nodes (this time the slope is exactly 1), since the tree degenerates to a linked list, as expected.

Random treaps, instead, keep performing well, with logarithmic height, just like for the other tests, even in the most unfavorable case.

We can then conclude that, if the parameter to improve is the height of the tree, randomized treaps do have an advantage in comparison to BSTs, and it is true that they keep a logarithmic height in all situations: therefore, their performance is comparable to more complex data structures like red-black trees.

The degeneration of BSTs in this test caused the recursive implementation of the `add` method, provided on our repo, to crash with a stack overflow[14] for n~=15K elements.